LAB 6

Code and Branch Coverage Analysis

Coupon Validator Application

Student Name	Mohammed Shahid Javed Mokashi
SRN	PES1UG24AM378
Course	Software Engineering Laboratory
Date	12 November 2026
Application Repository	L6_L7_31_coupon_validator

Submitted as evidence of the Lab 6 code coverage and branch coverage exercise.

1. Introduction

The objective of this lab is to understand the difference between code coverage and branch coverage, use the Python coverage tool to measure test completeness, interpret coverage reports, and improve test cases to exercise edge cases and decision branches.

The assigned application is a Coupon Validator. The lab requires separate test suites for code coverage and branch coverage, terminal evidence showing at least 90% coverage, and separate commits for the two updated test suites.

2. Application Overview

The application contains three functions. **coupon_discount()** validates coupon conditions and calculates a discount for WELCOME10, WEEKEND20, and FLAT100. **best_discount()** evaluates a list of coupons, selects the highest discount, and limits the result to 40% of the cart total. **is_expired()** determines whether a coupon has expired based on its validity period.

3. Methodology

The test suites were implemented using pytest, and coverage.py was used to collect statement and branch execution data. Tests were designed to exercise valid and invalid cart totals, new and existing users, weekend and weekday coupon conditions, cart-value thresholds, unknown coupon codes, discount selection and capping, and coupon expiry boundaries.

3.1 Code Coverage

Code coverage measures whether executable statements were run during testing. The code coverage suite contains 23 passing test cases. The terminal report records 32 statements, 0 missing statements, and 100% coverage.

3.2 Branch Coverage

Branch coverage measures whether decision outcomes were exercised. The branch suite tests both outcomes of the coupon conditions, the best-discount update and cap decisions, and expiry decisions. The terminal report records 21 passing test cases, 32 statements, 24 branches, 0 partial branches, and 100% coverage.

4. Results and Analysis

Metric	Code Coverage	Branch Coverage
Tests passed	23	21
Statements	32	32
Missing statements	0	0
Branches	Not reported	24
Partial branches	Not reported	0
Coverage	100%	100%

Both suites meet the lab's minimum requirement of 90% coverage and achieved 100% according to the supplied terminal output. The branch report also shows that all 24 measured branches were covered, with no partial branches.

5. Execution Commands

Code coverage:

```
py -m coverage erase
py -m coverage run --source=coupon_validator -m pytest -q
CodeCoverageTest_coupon_validator.py
py -m coverage report -m
```

Branch coverage:

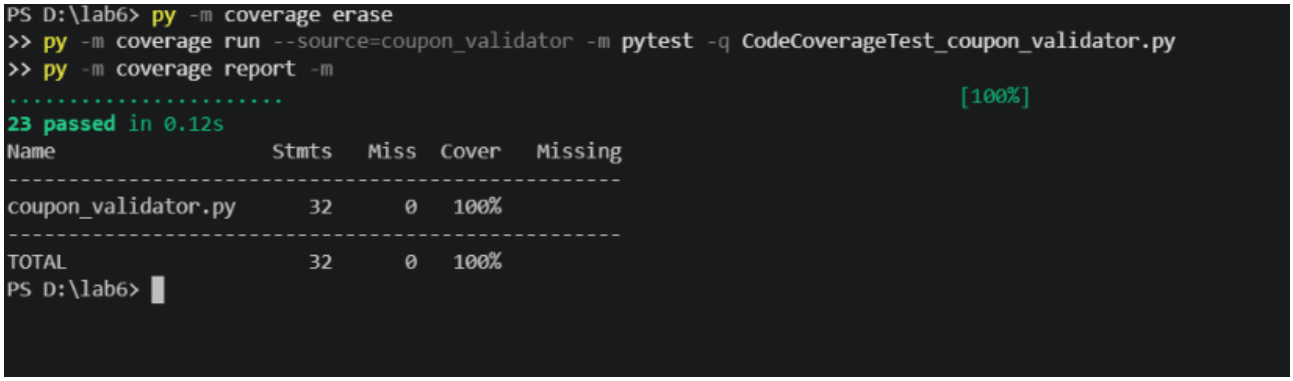
```
py -m coverage erase
py -m coverage run --branch --source=coupon_validator -m pytest -q
BranchCoverageTest_coupon_validator.py
py -m coverage report -m
```

6. Conclusion

The Coupon Validator test suites achieved 100% code coverage and 100% branch coverage in the supplied terminal runs. The exercise demonstrates that statement coverage and branch coverage provide complementary views of test completeness: statement coverage checks executed statements, while branch coverage checks alternative decision outcomes. The two test suites and terminal evidence are included with this report.

Appendix A — Code Coverage Terminal Evidence

The screenshot below shows 23 tests passed, 32 statements, 0 missing statements, and 100% total coverage.



Appendix B — Branch Coverage Terminal Evidence

The screenshot below shows 21 tests passed, 32 statements, 24 branches, 0 partial branches, and 100% total coverage.

